Contact

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Top Skills

Structural Design
Stress Analysis
MSC Nastran

Languages

Urdu (Native or Bilingual)
Punjabi (Elementary)
Saraiki (Native or Bilingual)
English (Professional Working)

Certifications

Introduction to Statistical Methods with MATLAB
“Non-linear modeling (using F.E.M)
Professional Engineer (PE)
PAT318A - Basic Durability and Fatigue Life Analysis Using MSC Fatigue
Fatigue Life & Damage Tolerance Analysis

Honors-Awards

Winner of Growww - The Challenge
1st Position in Mock Interview
South Asian Strategic Stability Institute (SASSI) Scholarship Program
Runner Up in Water Rocket Competition
Merit base Scholarship

Muhammad Shoaib Akram Sherani

Mechanical Engineer ✣ 5 UAVs, 2 Stores & 2 Suspension Systems with International Clients ✣ Project Engineer ✣ CAD / CAE / FEA Specialist ✣ I help Aerospace & Defense companies provide 100% reliable Structures ✣
Islamabad, Islāmābād, Pakistan

Summary

Mechanical Engineer with 10+ years of experience in the Aerospace & Defense industry. Proven track record of success in designing and developing novel products (MALE UAVs, Stores, Suspension Systems) that meet rigorous Aerospace standards (like STANAG 4671, FAR23, MIL-1530, MIL-8866, MIL-8591 & STP-1006). Skilled in CAD, FEM, Structural, Vibrational and Fatigue analysis & testing, data analysis & correlation, and technical documentation.

I thrive on the challenge of taking projects from concept to completion while collaborating with diverse stakeholders. Always seeking to learn and grow while implementing best engineering practices.

Let's Connect. Cheers.

Experience

NaqCoDE Technologies Pvt Ltd

Manager Structures
March 2024 - Present (1 year 3 months)
Islamabad, Islāmābād, Pakistan

Structure Design, Analysis, Prototyping & Testing of Aerospace products
Material Characterization as per ASTM standards including Composites
Collaboration with manufacturers
Timely manufacturing and delivery of products

Global Industrial & Defence Solutions (GIDS)

9 years 4 months

Manager (Project Engineer)

December 2020 - October 2024 (3 years 11 months)
Islamabad, Pakistan

Responsible for structural design, analysis and qualification of aerospace products in a timely and effective manner.

- Establish project plans with different stakeholders for well-defined deliverables
- Managing critical design reviews for higher management and different stakeholders
- Brainstorming and execution of design concepts in all phases of the Product Development Cycle using 3D CAD software, CATIA V5 & Creo Parametric.
- Structural configuration layouts and detail mechanical designs incl. preliminary sizing, joint layouts, interface boundary definitions of parts and assemblies for metallic and composite materials
- Drafting/Drawing generation with proper GD&T as per ASME Y14.5
- Development of metal & composite (GFRP & CFRP) structures finite element models for modal, buckling,
structural, and fatigue analyses to establish the required usage.
- Create, condense, and manage Global FEMs to optimize the design and performance of structures.
- Generate detailed testing instructions, instrumentation plan, and criteria for structural, fatigue,
and drop tests consistent with STANAG 4671, MIL-STD-1530, MIL-STD-8866 & STP-1006.
- Produce comprehensive technical reports that accurately document all technical aspects & findings

Structural Design & Analysis Engineer
July 2015 - November 2020 (5 years 5 months)
Islamabad, Pakistan

Working as Mechanical Structural Engineer and F&DT Engineer.

My responsibilities require effective communication with multiple project teams to gather scope of work, coordinate tasks and efficiently handling the project requirements for timely completion along with successful execution of engineering tasks.

- Structural configuration layouts and detail mechanical designs incl. preliminary sizing, joint layouts, interface boundary definitions of parts and assemblies for metallic and composite materials
- Drafting/Drawing generation with proper GD&T as per ASME Y14.5

- Performing Structural Analysis of critical components using MSC Patran, NASTRAN, Ansys WB and Abaqus for both metallic and composites (FRC) materials.
- Providing total life of components using MSC Fatigue (S-N, E-N) data base.
- Mapping the testing schemes through documentation to validate products against analyses.
- Execution of Static Testing for acceptance and qualification of critical components.
- Execution of Fatigue Testing at component level for required life hours.
- Performing post testing data analysis for validation of analysis.
- Documenting the analyses, testings and documentation through technical report writing.

Oil and Gas Development Company Limited, Pakistan
Trainee
December 2014 - January 2015 (2 months)
Logistic Base, Kot Sarang Field, Talagang. Pakistan

1) Visit to Rig N5 for
2) Participated in Maintenance of CAT Diesel Engine and Mud Pump
3) Cementing and Stimulation Process
4) Drilling process and Components (Drill Pipes, Drill Collars, Drill bits, Drilling Stabilizer)

M/s Trojans
Trainee Engineer
February 2014 - November 2014 (10 months)
Islamabad

Worked as Trainee Engineer with portfolio of Application, Sales and Design Engineer.
1) Delivered successful trainings of PTC - Creo Parametric Part, Assembly, Mechanism, Sheet metal and CAM modules to multiple including UK AID based program
2) Delivered multiple design concepts of different components using Creo Parametric
3) Provided services of Finite Element Modeling and Analysis as per clients requirements
4) Coordinated with multiple clients on different products (Ranging from 3D Modeling Packages, FEA Modules, CNC Machines, Reverse Engineering and Rapid Prototyping Products.)

5) Successfully integrated "Technical Education & Vocational Training Authority (TEVTA), Punjab" in Trojans Client list, which operates in 36 Districts of Punjab with an enrollment of 74,000 students per year

M/S TROJANS
Internship
July 2012 - July 2012 (1 month)

I worked as Internee at TROJANS Islamabad. I learned Creo Parametric, FARO ARM, Polyworks (Probing), Reverse engineering, Inspection and Quality assurance. I attended some Marketing sessions, where I learned how the organization work in the market.

Along with the above mentioned work portfolios, I participated in the different on-going projects at organization.

Education

Delft University of Technology
Non-Linear Modeling (using F.E.M) -(AE4ASM505, 3.0 ECTS, 84 Hours) - Online, Aerospace, Aeronautical and Astronautical Engineering · (January 2019 - April 2019)

Bahria University
Master of Science - MS, Project Management · (February 2018 - July 2019)

University of Engineering and Technology, Taxila
Bachelor of Science (B.Sc), Mechanical Engineering · (2010 - 2014)

Divisional Public School & College, D.G.Khan
Intermediate and Matriculation, Pre-Engineering and Science Group · (2003 - 2010)